ApproxDA-TransUNet: Understanding Global Context Sensitivity of Attention Approximation for Medical Image Segmentation

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***Abstract*—Dual-attention mechanisms have demonstrated strong performance in medical image segmentation by jointly modeling long-range spatial and channel dependencies, but their quadratic complexity has motivated numerous approximation methods whose task-dependent behavior remains poorly understood. To bridge this gap, we propose Approximate Dual Attention TransUNet (ApproxDA-TransUNet), an efficient dual-attention approximation framework for hybrid CNN-Transformer architectures that replaces global attention with controllable windowed low-rank spatial attention and grouped channel attention. Unlike existing efficient attention methods that primarily pursue computational savings, ApproxDA-TransUNet also provides a controllable framework for systematically studying how attention approximation affects segmentation performance. Extensive experiments on Synapse, Kvasir-SEG, and ISIC 2018 demonstrate consistent improvements over DA-TransUNet, achieving 80.94%, 90.17%, and 89.58% Dice, respectively, but also reveal two critical insights: First, Global Context Sensitivity (GCS)—tasks requiring long-range dependency are far more sensitive to approximation scale than texture-dominated ones; Second, Gate Collapse—the commonly adopted learnable attention gate consistently degrades into near-uniform routing due to gradient symmetry, undermining adaptive fusion. These findings challenge the prevailing view of approximation as a pure efficiency tool, re-framing it as a task-dependent inductive bias that must be carefully calibrated to dataset characteristics. Our code and analysis offer both a strong segmentation baseline and actionable principles for designing efficient medical transformers.**

***Index Terms*—medical image segmentation, dual attention, efficient transformers, attention approximation, global context sensitivity, gate collapse**

I. INTRODUCTION

Transformers have revolutionized medical image segmentation by enabling global long-range dependency modeling that convolutional networks cannot achieve. Dual attention [1]—combining a Position Attention Module (PAM) for spatial correlations and a Channel Attention Module (CAM) for channel correlations—has emerged as a powerful mechanism, and methods such as DA-TransUNet [2] demonstrated its effectiveness in hybrid CNN-Transformer architectures for multi-organ and polyp segmentation. The quadratic complexity of dual attention, however, has prompted a wave of efficient approximations—windowed attention [3], low-rank factorization [4], and grouped channel interaction [5]—that are now standard design choices. These strategies share an implicit assumption that approximation is universally applicable with minor accuracy trade-offs—an assumption rarely examined across tasks with fundamentally different spatial reasoning requirements:

When, and at what scale, is attention approximation beneficial for medical image segmentation?

To investigate this, we design **ApproxDA-TransUNet**[1]: a dual-attention architecture with three orthogonal, independently controllable approximation operators—windowed PAM, low-rank projection, and grouped CAM. ApproxDA-TransUNet serves as both a practical efficient model that improves over DA-TransUNet on all three benchmarks, and an interpretable experimental platform for studying when and why approximation helps.

Applying ApproxDA-TransUNet across three medical segmentation benchmarks reveals a structured pattern: the same windowed attention operator is highly window-sensitive on multi-organ CT yet largely window-robust on polyp and skin lesion segmentation. Systematic ablation confirms that appropriate window selection consistently benefits all three benchmarks, but the degree of sensitivity differs markedly across tasks with different spatial context requirements. We further find that learnable routing collapses to a fixed equilibrium due to gradient symmetry—a structural property of symmetric two-branch architectures, not a training artifact.

The main contributions of this paper are:

- **ApproxDA-TransUNet**: We propose ApproxDA-TransUNet, an efficient dual-attention approximation architecture with three orthogonal controllable axes—windowed PAM, low-rank projection, and grouped CAM—that consistently improves segmentation performance over the original DA-TransUNet and competitive Transformer-based baselines.

[1]Source code will be made available upon acceptance.

- **Empirical discovery of GCS**: Using ApproxDA-TransUNet as a controllable experimental platform, we present a systematic empirical study of attention approximation across heterogeneous medical segmentation tasks, uncovering a 3.6× difference in window sensitivity between Synapse and Kvasir-SEG. We term this task-level variability *Global Context Sensitivity (GCS)* and demonstrate that it predicts window-tuning importance across tasks.
- **Mechanistic insight**: We provide empirical evidence suggesting that attention approximation introduces a task-dependent inductive bias better aligned with tasks of lower contextual requirements, and further observe gradient symmetry collapse: learnable dual-branch routing degenerates to $g \approx 0.5$ regardless of window size—a structural limitation of symmetric two-branch gating.

The pattern we uncover is structured by GCS: tasks with high contextual requirements demand careful window tuning; those with low requirements are largely window-robust. The underlying mechanism is an inductive bias shift—approximation does not just reduce cost, it changes what the attention mechanism is sensitive to.

The remainder of this paper is organized as follows: Section II reviews related work; Section III presents the ApproxDA-TransUNet framework; Section IV reports results against state-of-the-art methods; Section V provides mechanistic analysis; Section VI concludes.

II. Related Work

A. CNN-based Medical Image Segmentation

Convolutional encoder-decoder architectures established the foundational paradigm for medical image segmentation, with U-Net [6] and its successors [7]–[10] progressively refining skip connections, attention gating, and multi-scale feature representations. Despite strong performance on local-feature-dominant tasks, their limited receptive fields hinder long-range anatomical reasoning.

B. Transformer-based Medical Image Segmentation

Transformer architectures have significantly advanced medical image segmentation by enabling global dependency modeling beyond the locality of convolutional neural networks. Representative methods such as TransUNet [11], Swin-Unet [12], UNETR [13], UCTransNet [14], TransNorm [15], and DAE-former [16] combine convolutional feature extraction with transformer-based contextual modeling to achieve state-of-the-art segmentation performance across a wide range of medical imaging tasks. These methods treat global context modeling as uniformly beneficial, however, without examining whether different tasks require different degrees of spatial context.

C. Dual Attention and DA-TransUNet

DANet [1] introduced Position Attention Modules (PAM) and Channel Attention Modules (CAM), capturing spatial and channel correlations at $\mathcal{O}(N^2C)$ and $\mathcal{O}(C^2N)$ complexity respectively. DA-TransUNet [2] adapted this design for medical image segmentation by inserting dual-attention blocks at multiple skip connections of a hybrid R50-ViT-B/16 encoder-decoder, achieving state-of-the-art accuracy while inheriting the quadratic cost of full-map attention. DBAANet [17] further extends dual-branch attention by aggregating position and channel attention across multiple scales, demonstrating the continued appeal of dual-attention paradigms for medical image segmentation.

D. Efficient Attention and Adaptive Routing

Windowed local attention [3], low-rank projection [4], and grouped channel interaction [5] have become standard strategies for reducing the cost of global attention, with learnable gating widely adopted to fuse multi-branch outputs [5], [16]. DAMamba [18] further shows that dual-attention routing transfers effectively to state-space models for lightweight breast ultrasound segmentation, confirming the versatility of multi-branch attention beyond standard transformer architectures. Existing efficient attention methods primarily optimize computational efficiency while implicitly assuming approximation is universally applicable. Whether approximation itself changes representation learning, and why it behaves differently across tasks with different contextual requirements, remains largely unexplored. This work addresses both questions by studying the context-sensitive behavior of approximation across heterogeneous segmentation tasks and exposing the gradient symmetry dynamics that cause adaptive routing to degenerate into fixed blending.

III. Method

This section introduces the overall architecture of ApproxDA-TransUNet and the placement of ApproxDABlocks within it, then details the three independently controllable approximation axes each block implements, and finally analyzes their computational complexity.

A. Architecture Overview

Figure 1 illustrates the overall architecture: ApproxDA-TransUNet adopts the hybrid CNN–ViT encoder-decoder structure of DA-TransUNet [2], with an R50-ViT-B/16 backbone pretrained on ImageNet-21K. Three CNN encoder stages produce multi-scale features at strides 4, 8, and 16; a 12-layer ViT processes 14×14 patch tokens. ApproxDABlocks replace all dual-attention modules at three skip connections—512 channels at 28×28, 256 at 56×56, and 64 at 112×112—and at the bottleneck. Each block routes activations through a Low-Rank Windowed PAM branch and a Grouped CAM branch, fused by a configurable gate g. Figure 2 details the internal structure of each ApproxDABlock and its three independently controllable approximation axes.

B. Attention Approximation Formulation

The original DA module performs dense spatial and channel attention over the full feature map, with quadratic complexity

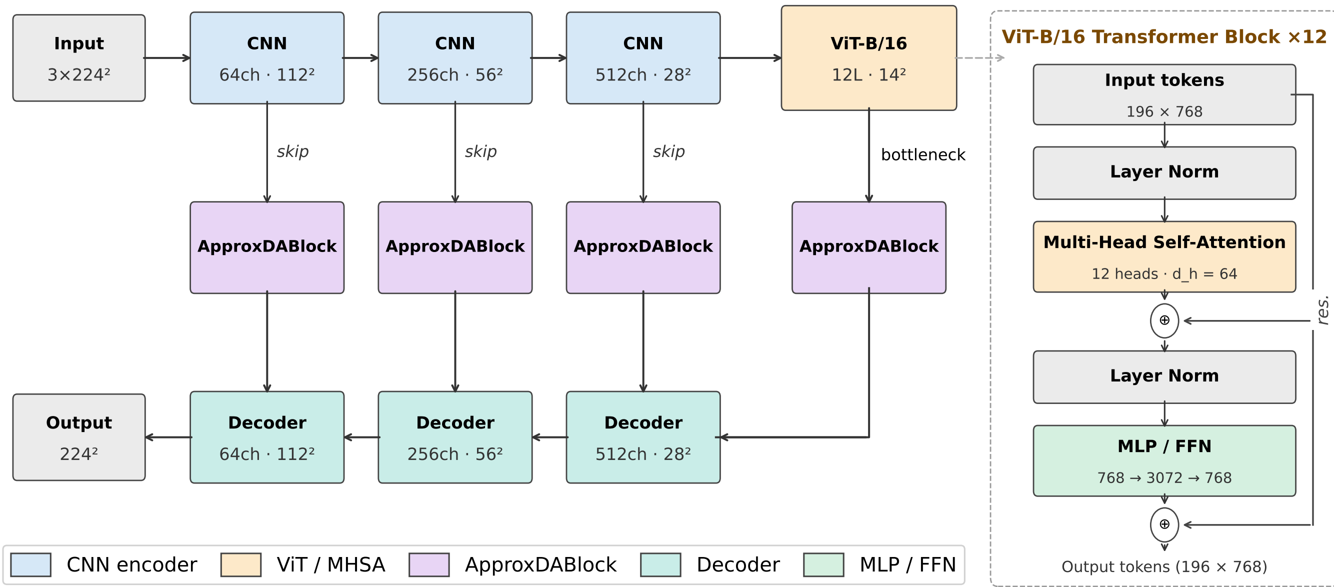

Fig. 1. ApproxDA-TransUNet architecture. ApproxDABlocks (Low-Rank Windowed PAM + Grouped CAM + gate routing) replace dual-attention modules at all skip connections and the bottleneck. `gate_mode` controls the routing during ablation.

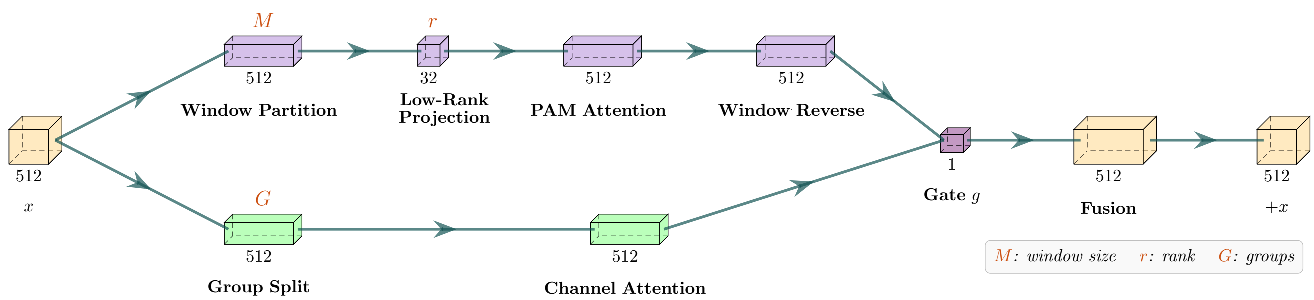

Fig. 2. ApproxDABlock internals with three independently controllable approximation axes: spatial scope (M, window size), representation fidelity (r, projection rank), and channel interaction (G, group count). Input $\mathbf{x}$ splits into a Low-Rank Windowed PAM branch (lavender, upper) and a Grouped CAM branch (green, lower); gate g fuses the outputs and $\mathbf{x}$ is added as a residual.

in both spatial resolution and channel dimension. ApproxDA-TransUNet reduces this cost along three independent axes, each targeting a distinct aspect of the approximation:

Spatial scope (window size M) controls *how far* PAM reaches. Full global attention ($M{=}H$) captures all long-range spatial relationships; restricting to $M{\times}M$ local windows trades long-range context for lower complexity. This axis most directly governs whether tasks with broader contextual requirements retain the anatomical context they depend on.

Representation fidelity (rank r) controls *how precisely* attention is computed within each window. Low-rank projection ($r \ll M^2$) approximates the affinity matrix at reduced cost; higher r approaches exact attention.

Channel interaction (groups G) controls *which channels* may interact in CAM. Partitioning C channels into G independent groups limits cross-channel dependency to within-group interactions, reducing the interaction scope.

These axes correspond to three distinct types of approximation: *spatial approximation* (how far attention reaches, controlled by M), *representation approximation* (how precisely affinity is computed, controlled by r), and *channel approximation* (how broadly channels interact, controlled by G). The axes are fully independent: any two of $\{M, r, G\}$ can be held fixed while the third varies, enabling clean ablations that isolate individual approximation effects.

C. Spatial Approximation

Instead of computing global position attention, ApproxDA-TransUNet partitions the feature map into non-overlapping windows of size $M{\times}M$. Attention is computed independently inside each local window, reducing complexity from $\mathcal{O}(N^2C)$ to $\mathcal{O}(NM^2C)$.

Window partitioning. Following [3], we partition the feature map into n_W non-overlapping $M\times M$ windows. Window clamping $M \leftarrow \min(M,H,W)$ allows M=112 to yield global attention at all scales, enabling a clean ablation that isolates windowing from low-rank as the performance bottleneck.

Low-rank projection. Following [4], query and key features inside each window are projected from N_w=M^2 to rank $r \ll M^2$ via a learned matrix $\mathbf{W}_r \in \mathbb{R}^{r\times N_w}$:

$$\tilde{\mathbf{C}} = \mathbf{W}_r\mathbf{C}^\top, \quad \tilde{\mathbf{D}} = \mathbf{W}_r\mathbf{D}^\top \in \mathbb{R}^{r\times C}. \tag{1}$$

The attention matrix $QK^\top \in \mathbb{R}^{N_w\times N_w}$ is approximated by $Q_rK_r^\top \in \mathbb{R}^{N_w\times r}$, where $Q_r, K_r \in \mathbb{R}^{N_w\times r}$, reducing total complexity to $\mathcal{O}(NrC)$. Combined, windowing and low-rank projection reduce the theoretical attention-matrix size from $\mathcal{O}(N^2)$ to $\mathcal{O}(Nr)$, a factor of N/r ($\sim$390$\times$ at 112$\times$112 with r=32). These two hyperparameters—window size M and rank r—form the primary experimental axes investigated on Synapse.

D. Channel Approximation

The original Channel Attention Module computes a $C\times C$ channel attention matrix at $\mathcal{O}(C^2N)$ cost. ApproxDA-TransUNet approximates this by partitioning C channels into G groups of width $C_g = C/G$, computing independent per-group attention:

$$X_{ji}^{(g)} = \frac{\exp(\mathbf{A}_i^{(g)} \cdot \mathbf{A}_j^{(g)})}{\sum_{k=1}^{C_g} \exp(\mathbf{A}_k^{(g)} \cdot \mathbf{A}_j^{(g)})}, \tag{2}$$

with total cost $\mathcal{O}(C^2N/G)$. Compared with dense channel interaction, grouped attention significantly reduces computational complexity while preserving local channel dependency.

The grouping number G provides another controllable approximation dimension, allowing the influence of channel approximation to be studied independently from spatial approximation.

E. Routing Strategy

The learnable routing mechanism of DA-TransUNet [2] is retained and extended to a configurable `gate_mode`. The outputs of the spatial and channel attention branches are fused via:

$$\mathbf{y} = \mathbf{W}_f\Big(g\cdot\hat{\mathbf{P}} + (1-g)\cdot\hat{\mathbf{C}}\Big) + \mathbf{x}, \tag{3}$$

where $\hat{\mathbf{P}}$ and $\hat{\mathbf{C}}$ are the normalized PAM and CAM outputs, $\mathbf{W}_f$ is a 1$\times$1 convolution, and g is determined by the `gate_mode` parameter:

`pam:`	$g = 1.0$	(PAM only)
`cam:`	$g = 0.0$	(CAM only)
`fixed:`	$g = 0.5$	(equal blend)
`learn:`	$g = \sigma(\mathbf{W}_g \cdot \mathrm{GAP}(\mathbf{x})) \in (0,1)^C$	

In gate=learn, $g \in (0,1)^C$ is a per-channel routing coefficient broadcast over all spatial positions; it collapses to a channel-wise blend rather than a spatially-adaptive or input-dependent scalar. This single-parameter design enables systematic ablation of routing strategy without any other architectural change. ApproxDA-TransUNet does not treat this routing module as a primary architectural contribution; rather, it serves as an analyzable component whose optimization dynamics—including an unexpected convergence to $g\approx0.5$ regardless of window size—are investigated in Section V.

F. Loss Function

The training objective combines Dice loss and cross-entropy loss with equal weights:

$$\mathcal{L} = \tfrac{1}{2}\mathcal{L}_{\text{Dice}} + \tfrac{1}{2}\mathcal{L}_{\text{CE}}. \tag{4}$$

IV. Results

We evaluate ApproxDA-TransUNet on three benchmarks spanning tasks with fundamentally different spatial context requirements, comparing against DA-TransUNet and other competitive baselines.

A. Experimental Setup

Datasets. We evaluate on three representative medical image segmentation benchmarks.

Synapse Multi-Organ CT [19]: 30 abdominal CT scans with an 18/12 train/test split covering 8 organ classes. Metrics: mean DSC and HD95.

Kvasir-SEG [20]: 1000 endoscopy polyp images (binary) with an 800/200 train/test split. Metrics: DSC and mIoU.

ISIC 2018 [21]: 2594 dermoscopy skin lesion images (binary) with a 2075/519 train/test split. Metrics: DSC and mIoU.

Implementation Details. All experiments use an R50-ViT-B/16 backbone pretrained on ImageNet-21K, trained on NVIDIA Tesla T4 (16 GB) GPUs. Optimizer: SGD, lr=0.01; batch size 24; input 224$\times$224; 300 epochs; best checkpoint selected by evaluating every 15 epochs. Default ApproxDA-TransUNet configuration: M=7, r=32, G=8.

Gate Protocol. ApproxDA-TransUNet supports four gate modes: `pam` (g=1.0, PAM only), `cam` (g=0.0, CAM only), `fixed` (g=0.5, equal blend), and `learn` (g learned per channel). Unless otherwise specified, all main-result experiments use `gate=pam`.

B. Comparison with State-of-the-Art

Tables I–III compare ApproxDA-TransUNet against published baselines across all three benchmarks.

Synapse. Table I benchmarks against 9 published methods. CNN-based models plateau at 76–78% DSC, with Att-UNet leading at 77.77%. Transformer-based methods progressively improve: TransUNet (77.48%), UCTransNet (78.23%), MIM (78.59%), Swin-Unet (79.13%), and DA-TransUNet (79.80%). ApproxDA-TransUNet with gate=pam, M=28, r=32 achieves **80.94%** DSC and 27.49 mm HD95—the highest DSC across all compared methods, surpassing DA-TransUNet by +1.14%. The intermediate window M=28 captures sufficient cross-organ context while limiting spurious long-range correlations; Section V analyzes this behavior in detail.

TABLE I SEGMENTATION RESULTS ON SYNAPSE MULTI-ORGAN CT

Method	DSC (%) ↑	HD95 (mm) ↓
CNN-based		
U-Net [6]	76.85	39.70
U-Net++ [7]	76.91	36.93
Residual U-Net [9]	76.95	38.44
MultiResUNet [10]	77.42	36.84
Att-Unet [8]	77.77	36.02
Transformer-based		
TransUNet [11]	77.48	31.69
UCTransNet [14]	78.23	26.75
TransNorm [15]	78.40	30.25
MIM [22]	78.59	26.59
Swin-Unet [12]	79.13	**21.55**
DA-TransUNet [2]	79.80	23.48
Dual-attention approximation (ours)		
ApproxDA-TransUNet (ours)	**80.94**	27.49

Kvasir-SEG. On polyp segmentation (Table II), U-Net leads the CNN-based group (88.22% DSC) while TransUNet (87.91%) and DA-TransUNet† (88.44%) are the strongest Transformer-based entries. ApproxDA-TransUNet with gate=pam, M=56 achieves **90.17%** DSC, **84.27%** mIoU, and **44.35 mm** HD95—state-of-the-art across all three metrics, with a $+1.73\%$ DSC gain and -8.69 mm HD95 improvement over the nearest baseline.

TABLE II SEGMENTATION RESULTS ON KVASIR-SEG POLYP

Method	DSC (%) ↑	mIoU (%) ↑	HD95 (mm) ↓
U-Net [6]	88.22	80.12	–
Att-Unet [8]	86.61	78.01	–
U-Net++ [7]	86.57	77.67	–
Res-UNet [9]	77.85	66.04	–
TransUNet [11]	87.91	80.03	–
DA-TransUNet [2]†	88.44	81.70	53.04
ApproxDA-TransUNet (ours)	**90.17**	**84.27**	**44.35**

†We are able to closely reproduce paper-reported results (88.47% DSC, 81.02% mIoU) and capture the HD95. ApproxDA: gate=pam, M=56, r=32.

ISIC 2018. On skin lesion segmentation (Table III), CNN-based methods cluster between 83–88% DSC, with TransUNet (88.78%) and DA-TransUNet (88.88%) leading among Transformer-based approaches. ApproxDA-TransUNet with gate=learn, M=7 achieves **89.58%** DSC (+0.70% over DA-TransUNet) and 82.66% mIoU, attaining the best DSC across all compared methods. This window-robust pattern is consistent with the low-GCS behavior observed on Kvasir-SEG, analyzed in Section V.

ApproxDA-TransUNet achieves state-of-the-art results across all three benchmarks (Figure 3), with task-dependent window sensitivity emerging as the key differentiating pattern across datasets.

TABLE III SEGMENTATION RESULTS ON ISIC 2018 SKIN LESION

Method	DSC (%) ↑	mIoU (%) ↑
U-Net [6]	87.22	81.14
Att-Unet [8]	87.60	81.51
U-Net++ [7]	87.30	81.33
Res-UNet [9]	83.32	76.51
TransUNet [11]	88.78	82.63
DA-TransUNet [2]	88.88	**82.78**
ApproxDA-TransUNet (ours)	**89.58**	82.66

ApproxDA: gate=learn, M=7, r=32, G=8.

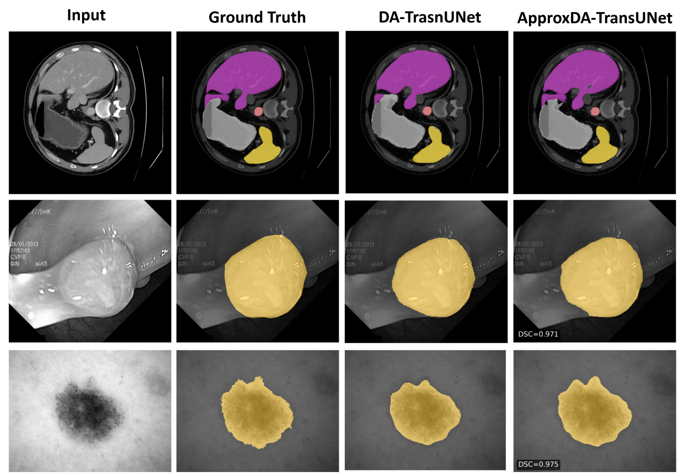

Fig. 3. Qualitative segmentation comparison. **Top (Synapse,** M**=28):** multi-organ CT; ApproxDA 80.94% DSC (+1.14%). **Middle (Kvasir-SEG,** M**=56):** polyp; 90.17% DSC (+1.73%). **Bottom (ISIC 2018,** M**=7):** skin lesion; 89.58% DSC (+0.70%).

C. Efficiency Analysis

Although the proposed approximation substantially reduces the theoretical complexity of the dual-attention module, the overall computational cost is dominated by the ViT backbone and convolutional encoder, which remain unchanged. Consequently, end-to-end GFLOPs and memory exhibit only marginal differences from DA-TransUNet. The practical advantage of ApproxDA-TransUNet therefore lies in improved architectural compatibility rather than total model complexity reduction: DA-TransUNet fails under DDP, while ApproxDA-TransUNet trains natively across multiple GPUs, enabling cost-effective distributed training on commodity clusters.

V. ABLATION STUDIES AND MECHANISTIC ANALYSIS

This section investigates why ApproxDA-TransUNet behaves differently under different approximation settings and segmentation tasks through a series of controlled ablation studies. Beyond validating the proposed design, these experiments provide mechanistic insights into attention approximation and its task-dependent behavior.

TABLE IV Efficiency Comparison on Synapse (NVIDIA Tesla T4)

Method	Params (M)	GFLOPs	Train VRAM	Train Time	Infer (s/vol)	Multi-GPU
DA-TransUNet [2]	107.95	30.2	11.5 G	11.4h	120.0	No†
ApproxDA-TransUNet (gate=pam, M=7)	112.98	32.1	6.4 G (×2 GPU)	8.3h (×2 GPU)	121.3	Yes (DDP)
ApproxDA-TransUNet (gate=learn, M=7)	114.90	32.1	10.6 G	11.5h	114.2	Yes (DDP)

†DA-TransUNet does not support multi-GPU training. ApproxDA-TransUNet uses Distributed Data Parallel (DDP); per-GPU batch 12, LR unchanged.

A. Approximation Strength Ablation

We first investigate how approximation strength influences segmentation performance by varying the spatial window size and approximation rank while keeping all other components unchanged. Table V isolates this effect by comparing $M \in \{7, 14, 28, 56, 112\}$ under gate=pam, r=32 on Synapse. Approximation is present in all rows—the right approximation *scale*, not the absence of approximation, governs performance.

TABLE V Spatial Approximation Ablation on Synapse (G=8)

Config	M	r	Gate	DSC (%) ↑	HD95 (mm) ↓
Windowed + low-rank	7	32	pam	78.64	31.09
Windowed + low-rank	14	64	learn†	78.04	29.09
Windowed + low-rank	28	32	pam	**80.94**	<u>27.49</u>
Windowed + low-rank	56	32	pam	78.90	35.23
Global + low-rank	112	32	pam	<u>79.44</u>	**27.26**

†Gate collapsed to $g\approx0.5$. M=112 via window clamping yields full-map attention at all scales.

The Synapse DSC-vs-M curve is *non-monotonic*: peak at M=28 (80.94%, +1.14% vs. DA-TransUNet), with all other scales underperforming—too-local (M=7: 78.64%), intermediate-upper (M=56: 78.90%), and fully-global (M=112: 79.44%). This pattern is analogous to the classical bias-variance tradeoff: under-context windows have high bias (missing necessary dependencies); over-context windows introduce spurious variance (irrelevant distant tokens); the intermediate M=28 achieves the optimal balance. The DSC range of 2.30 pp across M values confirms high window sensitivity on this task.

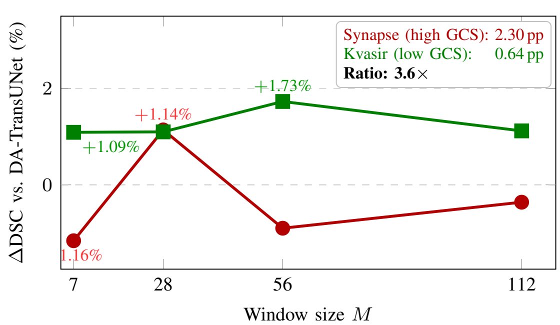

Fig. 4. ΔDSC vs. window size M (gate=pam, r=32) for Synapse (red, high GCS) and Kvasir-SEG (green, low GCS). Annotation box: DSC span and 3.6× ratio.

The M=14/r=64 row shows that increasing rank does not compensate for gate collapse ($g\approx0.5$, Section V-C): DSC 78.04% is lower than the gate=pam M=7/r=32 baseline (78.64%), suggesting r=32 already captures sufficient attention information and rank is not the performance bottleneck.

Preserving sufficient spatial receptive field is more important than increasing approximation rank; the optimal window scale—not approximation rank—governs segmentation performance.

B. Cross-task Ablation and Global Context Sensitivity

We next examine whether the same approximation strategy behaves consistently across different segmentation tasks. Although ApproxDA-TransUNet uses identical approximation operators across all experiments, its sensitivity to window selection varies substantially. On Synapse multi-organ CT, the DSC-vs-M curve (Figure 4) is non-monotonic with a sharp peak at M=28 and a 2.30 pp range across the measured window sizes. On Kvasir-SEG, the curve is nearly flat (0.64 pp range) with a shallow peak at M=56, indicating the task is largely window-robust. On ISIC 2018, ApproxDA-TransUNet (gate=learn, M=7) achieves 89.58% DSC vs. 88.88% for DA-TransUNet (+0.70%), consistent with the low-sensitivity pattern. With task-appropriate windows, gains are positive across all three benchmarks: +1.14% (Synapse, M=28), +1.73% (Kvasir, M=56), and +0.70% (ISIC, M=7).

These observations suggest that segmentation tasks exhibit different degrees of **Global Context Sensitivity (GCS)**—the degree to which accurate segmentation depends on preserving large-scale spatial context. Since GCS is not directly observable, we estimate it empirically through the performance response to window-size ablations, summarized by the DSC range:

$$\Delta\mathrm{DSC} = \max_M \mathrm{DSC}(M) - \min_M \mathrm{DSC}(M),$$

measured under fixed gate=pam, r=32. For Synapse, $\Delta\mathrm{DSC}$ = 2.30 pp; for Kvasir-SEG, 0.64 pp—a 3.6× ratio confirming that Synapse exhibits substantially higher empirical GCS. Tasks with higher empirical GCS exhibit steep DSC-vs-M curves with a well-defined optimal scale; lower-GCS tasks show broad performance plateaus where any reasonable window achieves competitive performance. Crucially, high GCS does not imply approximation is harmful: with the right window, ApproxDA-TransUNet outperforms full dual attention on *both* tasks. GCS characterizes how *sensitive* performance is to window choice—not the direction of the effect.

The task-level differences in GCS are grounded in the nature of each segmentation problem. Multi-organ abdominal CT segmentation requires reasoning over long-range anatomical relationships among multiple organs. The high sensitivity to window size reflects that both too-local windows (insufficient cross-organ context) and fully-global windows (noise from unrelated background) degrade performance; the intermediate scope (M=28) provides the optimal balance. In contrast, polyp and skin lesion segmentation are primarily governed by local appearance, boundary contrast, and texture consistency. For these low-GCS tasks, attention approximation introduces an *inductive bias* favoring local interactions while suppressing unnecessary long-range dependencies—a prior that better matches the intrinsic structure of the problem.

We validate this locality bias hypothesis via controlled attention-map analysis: comparing gate=pam M=7 (windowed) versus M=112 (global) attention effects on 10 Kvasir polyp images, windowed attention concentrates 62.3% of its effect on the polyp region versus only 34.2% for global attention—a $1.82\times$ difference confirmed in 9/10 samples (Figure 5). Windowed attention focuses on polyp boundaries while global attention disperses to surrounding tissue, providing direct evidence for the inductive bias alignment hypothesis.

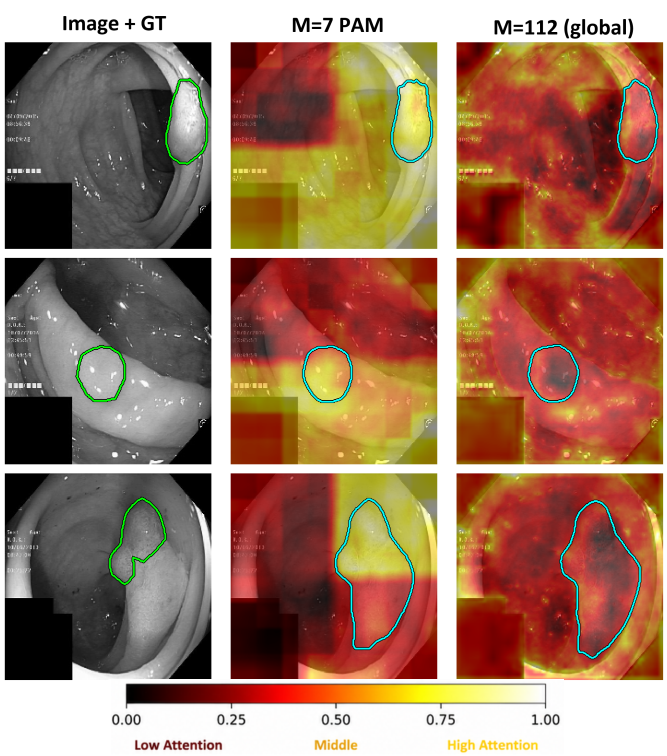

Fig. 5. PAM contribution maps ($\|\text{output} - \text{input}\|_2$, per-panel normalised; colorbar below) on three Kvasir-SEG polyp images. Each row: input with GT contour (left), windowed PAM (M=7; center), global PAM (M=112; right). Brighter color indicates larger PAM contribution at that location; panels are independently normalised so magnitudes are not comparable across panels.

Together, these findings reframe the role of approximation: not as a capability reduction but as a task-dependent inductive bias adjustment whose value depends on how well the enforced locality prior aligns with the task's intrinsic context scale.

C. Gate Ablation and Mechanistic Analysis

Finally, we investigate whether adaptive routing contributes to approximation performance by comparing PAM-only, CAM-only, and learnable gate configurations. Table VI reports results for all four gate modes on Synapse (M=7, r=32).

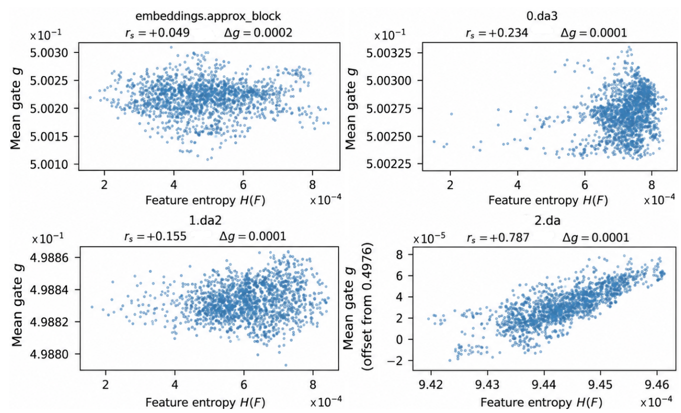

Fig. 6. Gate value g vs. per-block feature entropy $H(F)$ (M=7, gate=learn). Top-left: bottleneck; top-right: 28^2 skip; bottom-left: 56^2 skip; bottom-right: 112^2 skip. All four locations cluster at $g{\approx}0.5$ with $\Delta g{\leq}0.0002$.

TABLE VI GATE MODE ABLATION ON SYNAPSE (M=7, r=32 UNLESS NOTED)

Gate Mode	g	DSC (%) ↑	HD95 (mm) ↓
gate=pam	1.0	**78.64**	31.09
gate=cam	0.0	<u>78.26</u>	<u>30.59</u>
gate=learn (M=14, r=64)	$\approx$0.5*	78.04	**29.09**
gate=learn	$\approx$0.5*	77.78	34.29

*Gate collapsed to $g{\approx}0.5$ throughout training.

Spatial attention alone provides stronger routing than either channel-dominated or learnable fusion. Gate=pam (78.64%) outperforms gate=cam (78.26%) and both gate=learn configurations (77.78% at M=7; 78.04% at M=14, r=64). Notably, gate=learn with a larger window and higher rank (M=14, r=64) still underperforms gate=pam at M=7 (78.04% vs. 78.64%), confirming that gate collapse is more damaging than a suboptimal window choice. In both gate=learn configurations, the gate converges to $g{\approx}0.5$ regardless of window size, producing a fixed 50/50 blend that performs below both single-branch modes.

The collapse is not isolated to one location but is uniform across the entire architecture. Figure 6 provides direct evidence across all four ApproxDABlock locations (top-left: bottleneck; top-right: 28^2 skip; bottom-left: 56^2 skip; bottom-right: 112^2 skip): gate values show no correlation with feature entropy (Spearman $r_s{\approx}0$), ruling out sample-adaptive routing. This collapse is identical at both M=7 and M=14, confirming it is a fundamental property of symmetric two-branch gating rather than a windowing artifact.

Optimization intuition. Given fused output $\mathbf{F} = g\mathbf{F}_{\mathrm{P}} + (1-g)\mathbf{F}_{\mathrm{C}}$, the gate gradient is:

$$\frac{\partial \mathcal{L}}{\partial g} = \frac{\partial \mathcal{L}}{\partial \mathbf{F}} \cdot (\mathbf{F}_{\mathrm{P}} - \mathbf{F}_{\mathrm{C}}) . \tag{5}$$

At initialization $g{\approx}0.5$, both branches receive equal gradient $\frac{1}{2}\partial\mathcal{L}/\partial\mathbf{F}$. Without incentive to differentiate, $\mathbf{F}_{\mathrm{P}} \approx \mathbf{F}_{\mathrm{C}}$, so Eq. (5)$\approx$ $\mathbf{0}$. This is self-reinforcing: equal gradients $\rightarrow$ convergent representations $\rightarrow$ near-zero gate gradient $\rightarrow$ g fixed at 0.5—a stable equilibrium of symmetric dual-branch optimization.

Empirical evidence. Quantitative gate monitoring confirms this collapse is near-total across all window configurations. After 300 epochs, the gate range at $M{=}7$ is $[0.4993, 0.5010]$ with $\Delta g{=}0.0017$. At $M{=}14$: $g{\approx}0.5002$ with $\Delta g{\approx}0.0000$.

Simple scalar gating cannot provide effective adaptive routing because branch symmetry prevents meaningful specialization. Future adaptive attention mechanisms should explicitly encourage branch diversity—through asymmetric initialization, diversity-encouraging loss terms, or non-symmetric routing—rather than relying solely on learnable fusion weights.

VI. Conclusion

This paper presented ApproxDA-TransUNet, an efficient dual-attention approximation framework for medical image segmentation. Across three representative benchmarks, ApproxDA-TransUNet consistently outperformed the original DA-TransUNet while retaining efficient attention computation, demonstrating that appropriately designed attention approximation can improve segmentation accuracy rather than merely reducing computational cost. Beyond the proposed architecture, our experiments provide new insights into attention approximation. We show that approximation effectiveness depends on the contextual requirements of the target task, which we describe as Global Context Sensitivity (GCS). High-GCS tasks require careful approximation design, whereas low-GCS tasks remain robust across a broad range of approximation scales. We further demonstrate that conventional scalar gate learning collapses toward nearly uniform routing because of gradient symmetry, suggesting that effective adaptive fusion requires mechanisms that explicitly promote branch specialization. Overall, our results indicate that attention approximation should be regarded as a task-dependent inductive bias instead of solely an efficiency technique. GCS currently remains an empirical characterization: estimating task GCS without window-size ablations is an open problem, and the inductive-bias framing warrants further theoretical grounding. We hope this work motivates future research toward principled, task-aware design of efficient attention mechanisms for medical image segmentation.

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